

Sniper Trader — The Master Manual



The Grandmaster: A cosmic entity playing a complex game of 3D chess, symbolizing the meta-optimizer managing portfolios.

Prologue: The Philosophy of the Sniper

The market is not a random walk; it is a mechanism of wealth transfer designed to extract liquidity from the impatient and deliver it to the disciplined. The **Sniper Trader** system is built on a single, uncompromising philosophy: **Protect Capital at All Costs.**

We do not predict the market. We mathematically trap it.

To achieve this, the system operates as a **3-Bot Framework**, where three distinct archetypes—Mage, Sage, and Seer—hunt specific market inefficiencies. They are orchestrated by the **Grandmaster**, a meta-optimizer that constructs the Holy Grail portfolio through Deterministic Clustered Risk Parity (DCRP).

Part I: System Architecture

The entire ecosystem is strictly divided into two computational tiers.

The Golden Parity Rule

Any change to a trading rule (entries, trailing stops, break-even, spread filters, body filters) **MUST be applied symmetrically across BOTH tiers simultaneously**.

```
graph TD
  subgraph Tier 1: Alpha Generation (Math / Backtest)
    A1[Optimizer Worker Threads]
    A2[Walk-Forward Engine]
    A3[Mage/Sage Math Core]
  end
  subgraph Tier 2: Live Production (Orchestrator)
    B1[MetaApi TickFeed]
    B2[LiveOrchestrator.ts]
    B3[Shadow Backtester Parity Bridge]
  end
  A3 <-->|Strict 0.01R Tolerance| B3
```

- **Tier 1 (The Laboratory):** This is where alphas are born. Millions of combinations are simulated in seconds using the highly optimized `MathCore` arrays.
 - **Tier 2 (The Battlefield):** This is the live event-driven engine (`LiveOrchestrator.ts`) that listens to real-time broker ticks and executes the exact logic simulated in Tier 1.
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Part II: The 3-Bot Framework

1. MAGE: The Open Range Breakout (ORB) Hunter



The Mage: Breaking through boundaries and capturing explosive momentum.

Mage is designed to capture structural momentum shifts at the onset of major trading sessions (Asia, London, New York).

The Mechanics:

The Build Phase: Mage waits for a specific time (e.g., 3:00 AM EST) and begins recording the highest high (`orHigh`) and lowest low (`orLow`) over a defined window (e.g., 30 minutes). This creates the "box".

The Breakout: Mage stalks the box for up to 4 hours. If an M5 candle closes outside the box with a body larger than `minBodyPips`, the trap is sprung.

Execution: A market order is fired instantly. The Stop Loss is placed on the opposite side of the box (plus spread).

The Trail (ADTEL): If the trade moves in profit, Mage ruthlessly trails the price, locking in gains tick-by-tick.

2. SAGE: The Liquidity Sweep Fader



The Sage: Reversing momentum, turning the market's violence into calm, flowing water.

Sage is the yin to Mage's yang. While Mage buys breakouts, Sage assumes the breakout is a lie—a **liquidity hunt** by institutional algorithms to trap retail traders.

The Mechanics:

The Sweep: Sage watches the exact same ORB box as Mage. If price aggressively pokes outside the box by `sweepPips` but fails to hold, Sage goes on high alert.

The Rejection: The `actionCandle` (a 5, 10, or 15-minute timeframe) must close **back inside** the box. This confirms the breakout was a fakeout. Furthermore, the candle must exhibit a Wick-to-Body Ratio (WBR) of at least 1.5.

The Limit Trap: Sage does not chase the price. It calmly places a Limit Order at the edge of the box and waits for price to retrace and fill it.

The Reversal: Once filled, Sage rides the reversal back to the midpoint or the opposite boundary.

```
sequenceDiagram
    participant Market
    participant SageEngine
    participant Broker
    Market->>SageEngine: Price sweeps ORB High
    Market->>SageEngine: Candle closes back inside (WBR > 1.5)
    SageEngine->>Broker: Place Limit Sell at ORB High
    Market->>Broker: Price pulls back (Limit Filled)
    Broker-->>SageEngine: Position Opened
    SageEngine->>Broker: Trail Stop Loss downward tick-by-tick
```

3. SEER: The Visionary Intelligence



The Seer: Peering through the fabric of reality using Vision AI.

Seer brings artificial intelligence directly into the execution loop. It is modeled after the Stacy Burke liquidity hunt methodology.

The Mechanics:

Mathematical Candidates: Seer constantly scans the market for structural setups—Day 3 breakouts, First Red Days, Inside Days.

Chart Rendering: When a candidate is found, Seer uses the `ChartRenderer` to draw a perfect PNG image of the chart, complete with EMAs, session boundaries, and liquidity levels.

Vision Evaluation: The image is sent to the Gemini Vision API alongside the `StacyBurkePrompt`. The AI acts as a seasoned discretionary trader, evaluating market structure, time-of-day, and trap potential.

Execution: If the AI approves with high confidence, the trade is routed to the broker.

Part III: Defensive Engineering

The Margin Defender (`tradeManager.ts`)

Operating on a separate polling loop, this is the ultimate safety net. If account equity drops below safe margin levels, it executes emergency liquidation.

Daily Drawdown Circuit Breaker (DWCB)

Calculated via `DwcbCalculator.ts`, if an account violates a 4.5% daily drawdown, `GlobalTradeGate` shuts down all new entries system-wide until midnight EST.

The News Death Zone

Five minutes before High-Impact News (NFP, CPI, FOMC) and during the 16:50 EST broker rollover, the `LiveOrchestrator` sweeps the portfolio and cancels all pending orders to prevent slippage annihilation.

Part IV: The Grandmaster Synthesis

The final stage of the Sniper Trader system is not trading—it is **Portfolio Synthesis**.

The optimizers (`mage_optimizer.ts`, `sage_optimizer.ts`) run massive Walk-Forward Analysis (WFA) simulations over years of tick data, dumping raw alphas into the `optimizer_dump/` directories.

The `GrandmasterGA.ts` then takes over. It uses Combinatorial Purged Cross-Validation (CPCV) and Probabilistic Lookahead-free WFO (PLWFO) to assemble a portfolio that mathematically caps the Maximum Drawdown at exactly 1.00R (99% Monte Carlo confidence) while concentrating risk capital into the highest Sharpe ratio, low-correlation components.

This is Deterministic Clustered Risk Parity (DCRP). This is the Holy Grail.